

1. Remove Element

Given an integer array `nums` and an integer `val`, remove all occurrences of `val` in `nums` **in-place**. The order of the elements may be changed. Then return *the number of elements in `nums` which are not equal to `val`*. Consider the number of elements in `nums` which are not equal to `val` be `k`, to get accepted, you need to do the following things:

- Change the array `nums` such that the first `k` elements of `nums` contain the elements which are not equal to `val`. The remaining elements of `nums` are not important as well as the size of `nums`.
- Return `k`.

Custom Judge:

The judge will test your solution with the following code:

```
int[] nums = [...]; // Input array
int val = ...; // Value to remove
int[] expectedNums = [...]; // The expected answer with correct length.
// It is sorted with no values equaling val.
```

```
int k = removeElement(nums, val); // Calls your implementation
```

```
assert k == expectedNums.length;
sort(nums, 0, k); // Sort the first k elements of nums
for (int i = 0; i < actualLength; i++) {
    assert nums[i] == expectedNums[i];
}
```

If all assertions pass, then your solution will be **accepted**.

Example 1:

Input: `nums = [3,2,2,3]`, `val = 3`

Output: 2, `nums = [2,2,_,_]`

Explanation: Your function should return `k = 2`, with the first two elements of `nums` being 2. It does not matter what you leave beyond the returned `k` (hence they are underscores).

Example 2:

Input: `nums = [0,1,2,2,3,0,4,2]`, `val = 2`

Output: 5, `nums = [0,1,4,0,3,_,_,_]`

Explanation: Your function should return `k = 5`, with the first five elements of `nums` containing 0, 0, 1, 3, and 4.

Note that the five elements can be returned in any order.

It does not matter what you leave beyond the returned `k` (hence they are underscores).

Constraints:

- $0 \leq \text{nums.length} \leq 100$
- $0 \leq \text{nums}[i] \leq 50$
- $0 \leq \text{val} \leq 100$

PROGRAM

```
class Solution {  
  
    public int removeElement(int[] nums, int val) {  
  
        int k = 0;  
  
        for (int i = 0; i < nums.length; i++) {  
  
            if (nums[i] != val) {  
  
                nums[k] = nums[i];  
  
                k++;  
  
            }  
  
        }  
  
        return k;  
  
    }  
}
```

OUTPUT

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VTU30221 submitted at Sep 12, 2026 19:25

Analysis Solution

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Runtime

0 ms | Beats 100.00%

Memory

43.33 MB | Beats 78.71%

150%

100%

50%

0%

1ms 2ms 3ms 4ms

Code | Java

1 class Solution {

7

8

9

10

11

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13

14

15

16

if (nums[i] != val) {

nums[k] = nums[i];

k++;

}

return k;

}

Saved Ln 16, Col 2

Testcase > Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

nums =

[3,2,2,3]

val =

3

Output

31°C Mostly cloudy

Search

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